

Process Characterization Plan

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCP-007

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Approvals	3
Abbreviations	3
1 Purpose and scope	5
2 Objectives	6
3 Related documents	7
4 Prior knowledge and quality risk basis	8
4.1 Unit-operation description and prior knowledge	8
4.2 Quality attributes in scope	8
4.3 Risk-based prioritization of parameters	9
5 Materials and methods	10
5.1 Scale-down model and its qualification	10
5.2 Operation	10
5.3 Analytical methods	10
5.4 Sampling plan	11
5.5 Statistical methods	11
6 Study design	12
6.1 Factors, ranges and study type	12
6.2 Screening design	12
6.3 Response-surface design	13
6.4 Univariate assessment	13
7 Acceptance and decision criteria	14
8 Data management and integrity	15
9 Roles and responsibilities	16
10 Deliverables and schedule	17
11 Risks and assumptions	18
12 Approvals	19
References	20
Appendix A — Planned screening design	21

Field	Value
Document ID	PCP-007
Document class	Process Characterization Plan
Title	Process Characterization Plan: Cation Exchange Chromatography (Step 7)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion-exchange chromatography · ANOVA, analysis of variance · CCD, central-composite design · CEX, cation-exchange chromatography · CPP, critical process parameter · CQA, critical quality attribute · CV, column volume · DoE, design of experiments · DS, drug substance · ELISA, enzyme-linked immunosorbent assay · GPP, general process parameter · HCP, host-cell protein · HMW, high-molecular-weight (aggregate) · ICH, International Council for Harmonisation · KPP, key process parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · QbD,

Quality by Design · RRF, risk ranking and filtering · RSM, response-surface methodology · SDM, scale-down model · SEC, size-exclusion chromatography · WC-CPP, well-controlled CPP.

1 Purpose and scope

This Process Characterization Plan (**PCP-007**) defines the Stage 1 (Process Design) characterization of the A-Mab cation-exchange (CEX) polishing chromatography step (Step 7), the bind-and-elute polishing operation that follows low-pH viral inactivation. CEX is the **principal aggregate-reduction (polish) step** of the purification train — the only step that actively reduces high-molecular-weight aggregate — and provides a major further reduction of host-cell protein (HCP), and modular clearance of residual DNA and leached Protein A. The plan specifies the objectives, the attributes in scope, the process parameters and ranges to be characterized, the experimental designs, the analytical methods, the sampling plan, the statistical-analysis approach, and the acceptance and decision criteria by which each parameter will be classified and a multivariate operating region established.

Scope. The commercial-scale cation-exchange polishing step, characterized on a qualified scale-down chromatography column. The upstream low-pH viral-inactivation step (Step 6), which defines the feed to this step, and the downstream anion-exchange step (Step 8), which provides the subsequent polish, are addressed only where they define the feed to, or clear the impurity load from, this step; their characterization is out of scope. This plan governs execution only — the results are reported in the paired Process Characterization Report (**PCR-007**).

Basis. The unit-operation model, quality-attribute framework and risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009); the Stage 1 approach and report structure follow PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the QbD framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012) and the lifecycle approach of the FDA 2011 guidance (U.S. Food and Drug Administration 2011).

2 Objectives

The characterization will:

1. establish the quantitative relationships between the cation-exchange process parameters and the eluate-pool aggregate, the eluate-pool HCP and the step yield;
2. identify the significant main effects, two-factor interactions and curvature, including the anticipated protein-load $\times$ wash-conductivity interaction that governs HCP clearance;
3. define the multivariate operating region over which the eluate-pool aggregate and HCP presented to the anion-exchange step are controlled;
4. establish proven acceptable ranges (PARs) and confirm normal operating ranges (NORs) for each parameter;
5. classify each parameter for its risk of impact to CQAs and to process performance (CPP / WC-CPP / KPP / GPP); and
6. provide the process understanding and ranges that justify the Stage 2 operating conditions and the drug-substance control strategy.

3 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 3.1: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-007	Process Characterization Report (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3.2 (numbers are placeholders in this synthetic set).

Table 3.2: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Cation-exchange chromatography is the first polishing step of the A-Mab purification train: the neutralized viral-inactivation pool is loaded onto a strong cation-exchange resin at controlled conductivity, weakly bound impurities are removed in the wash, and the product is recovered by an ascending-pH (or ascending-salt) elution, with the eluate pool collected between defined start and stop criteria (CMC Biotech Working Group 2009). Unlike the upstream operations it forms no new product-quality CQA; its role in the control strategy is to **reduce the high-molecular-weight aggregate** carried into the pool — including the aggregate that can arise during the upstream low-pH hold — and to provide a major further reduction of HCP and modular clearance of residual DNA and leached Protein A. Platform knowledge established that aggregate, being more highly charged and more strongly retained, is separated from monomer in the elution so that the pool aggregate rises with higher protein load and as the elution pool is extended further down the descending edge (a higher stop-collect) and at higher elution pH; that HCP clearance is governed by the load/wash conductivity — a higher conductivity strips more weakly bound HCP before elution — and falls at high load, the two acting together through a load $\times$ conductivity interaction; and that recovery declines modestly at high load. This knowledge frames the attributes in scope and the prioritization of parameters for characterization.

4.2 Quality attributes in scope

Cation exchange introduces no CQA. It is the principal reduction step for the high-molecular-weight aggregate and a major clearance step for the impurity CQAs formed upstream and specified at the drug substance (Table 4.1). Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact $\times$ Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009; International Council for Harmonisation 2023).

Table 4.1: CQAs controlled by the cation-exchange step (formed upstream; polished/cleared here), with acceptance criterion and criticality (Tool #1 = Impact $\times$ Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

The eluate-pool aggregate and eluate-pool HCP are not themselves released specifications; they are in-process attributes whose control at this step determines the size-variant and impurity load presented to the anion-exchange step and therefore the margin to the 5% HMW aggregate and 100 ng/mg HCP drug-substance limits. Because aggregate can be elevated in the incoming pool, the study is designed to demonstrate the aggregate-clearance capacity of the step against a representative, deliberately challenging aggregate feed; pool aggregate is the primary response of the study.

4.3 Risk-based prioritization of parameters

Consistent with the A-Mab lifecycle of iterative risk assessments and the risk flow of PDA TR 60, the study scope is set by the Pre-Characterization Process Risk Assessment (**RA-001**). A risk-ranking-and-filtering (RRF) screen scores each parameter for its potential main-effect and interaction impact and assigns the study type accordingly: protein load, load/wash conductivity, elution pH and the elution stop-collect — which carry a credible main-effect and interaction risk to the eluate-pool aggregate or HCP — are studied in the multivariate DoE; the elution flow rate, expected to affect only process performance over a wide range, is studied univariately (CMC Biotech Working Group 2009). The resulting parameter set and study-type assignment are given in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

Characterization will be performed on a scale-down cation-exchange column operated at the commercial-process equivalents for resin, bed height, residence time, buffer system, load ratio and elution strategy (**SOP-2010**, **SOP-2008**). Prior to characterization the SDM will be qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and, critically, the input and output attributes — step yield, eluate-pool aggregate and HCP, residual DNA and leached Protein A — to confirm the model is representative and suitable to support operating-range and PAR claims (Parenteral Drug Association 2013). The centre-point replicates of each design provide an in-study estimate of model reproducibility (pure error) used in the lack-of-fit assessment.

5.2 Operation

The neutralized viral-inactivation pool is conditioned to the target load conductivity and loaded onto the equilibrated cation-exchange column to the target load ratio; the column is washed at the load/wash conductivity to remove weakly bound host-cell protein and other impurities; and the product is recovered by elution, with the eluate pool collected from a defined start until the descending-edge stop-collect criterion. Buffers, resin lot and column packing are controlled to platform specifications (**SOP-2008**) and are treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009).

5.3 Analytical methods

The step responses will be measured by the validated methods in Table 5.1; method performance characteristics are per the cited validation reports. Aggregate is determined by SEC-HPLC, the primary assay for the aggregate-polish claim.

Table 5.1: Validated analytical methods for the characterization.

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

5.4 Sampling plan

Each DoE run will be sampled at the eluate pool for aggregate, HCP, residual DNA and leached Protein A, and the load and pool product mass will be recorded for the step-yield determination; in-process readings (pH, conductivity, UV) will confirm the chromatographic profile and the pool collection window. Centre-point runs are replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

5.5 Statistical methods

Designs will be constructed and analyzed in coded factors, with each parameter's characterization range mapped so that the set-point is 0 and the range edges are ± 1 (**SOP-1002**). Screening data will be analyzed by ordinary-least-squares regression of each response on the main effects and all two-factor interactions; factor effects are reported as twice the coded coefficient. Response-surface data will be analyzed with the full quadratic model (main effects, two-factor interactions and pure-quadratic terms). Significance will be assessed at $\alpha = 0.05$. Model adequacy will be judged by R^2 , adjusted R^2 , predicted R^2 (from the PRESS statistic), the overall model F-test, and an ANOVA lack-of-fit test against the centre-point pure error; a response with no significant parameter effect beyond a single dominant factor will be reported as robust rather than modelled predictively. The operating region will be defined as the multivariate region of the well-controlled CPPs over which the eluate-pool aggregate and HCP are controlled, and commercial-scale capability of the drug-substance CQAs governed by the step will be estimated by Monte-Carlo simulation with each parameter varied within its NOR.

6 Study design

6.1 Factors, ranges and study type

Five parameters are in scope (Table 6.1). The characterization range of each factor (“Range studied”) defines the edges of the explored knowledge space and the basis of the PAR; the NOR is the tighter routine-control range. Classification is an output of the study and is reported in PCR-007. The DoE factor coding for the four multivariate factors is given in Table 6.2.

Table 6.1: Cation-exchange parameters, set-points, characterization ranges and planned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	25	10–40	18–32	multivariate
Load/Wash conductivity	mS/cm	5	3–7	4–6	multivariate
Elution buffer pH	pH	6	5.8–6.2	5.85–6.15	multivariate
Elution stop collect	OD desc.	1	0.5–1.5	0.7–1.3	multivariate
Elution flow rate	cm/hr	200	100–300	150–250	univariate

Table 6.2: Design-of-experiments factor legend (coded A–D).

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

6.2 Screening design

Factor screening will use a two-level full factorial in the four multivariate factors (A–D) with three centre-point replicates, 19 runs in total. The full factorial estimates all four main effects and all six two-factor interactions without confounding. Three responses will be recorded for each run (eluate-pool aggregate, eluate-pool HCP and step yield). The planned coded design is given in Appendix A.

6.3 Response-surface design

The four multivariate factors will be carried into a face-centred central-composite design with four centre-point replicates, 28 runs in total, enabling estimation of curvature in addition to main effects and interactions. The planned coded design is given in Appendix B.

6.4 Univariate assessment

The elution flow rate will be evaluated one factor at a time across its characterization range for its effect on step yield, the eluate-pool aggregate and HCP, and the chromatographic profile; it will not be carried into the multivariate design because prior knowledge indicates no credible impact to a product-quality CQA.

7 Acceptance and decision criteria

The study is acceptable when:

1. the SDM qualification confirms representativeness of commercial scale (**SOP-1001**);
2. the fitted response-surface models for the eluate-pool aggregate and the eluate-pool HCP are adequate — target $R^2 = 0.90$, adjusted $R^2 = 0.80$, no significant lack of fit ($p > 0.05$ against the centre-point pure error), and residual diagnostics without systematic structure (a response governed by a single dominant factor is reported by that dependence and is otherwise robust); and
3. a multivariate operating region exists over which the eluate-pool aggregate and HCP are controlled at the set-point and across each parameter's NOR.

Each parameter will then be classified per the A-Mab continuum (CMC Biotech Working Group 2009):

- a parameter whose variability significantly impacts a CQA is a **CPP**; where the impact is reliably controlled within the operating region (low risk of leaving it) the parameter is a **well-controlled CPP (WC-CPP)**;
- a parameter that does not significantly impact a CQA but affects process performance or consistency is a **KPP**; otherwise it is a **GPP**;
- consistent with the post-characterization FMEA convention, a parameter that affects a CQA with Severity 8, or that carries an initial (pre-mitigation) risk priority number > 72 , is designated a CPP before mitigation by the control strategy.

The operating region, classifications and PARs established here will justify the Stage 2 operating ranges and feed the drug-substance control strategy.

8 Data management and integrity

All raw data, DoE designs, analytical results and statistical analyses will be recorded and retained under the site data-integrity procedures (ALCOA+). Analyses will be performed with version-controlled scripts and reviewed by a second analyst; designs, models and the derived operating region will be archived with the report. Any deviation from this plan will be documented and assessed for impact in PCR-007.

9 Roles and responsibilities

Table 9.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the DoE; author PCR-007
Analytical Development	Perform released and in-process testing under the referenced AMV methods
Manufacturing Science	Qualify the scale-down model; confirm commercial-scale representativeness
Statistics	Review the design, the statistical analysis and the operating-region definition
Quality Assurance	Approve the plan and the report; confirm parameter classifications

10 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report (**PCR-007**), reporting the executed designs, the significant effects, the operating region, the parameter classification and the aggregate/impurity-CQA outcomes. PCR-007 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment. Execution follows SDM qualification and precedes the Stage 2 PPQ campaign.

11 Risks and assumptions

The plan assumes the SDM is representative of commercial scale (confirmed by qualification) and that the platform analytical methods are validated and in control. The principal risk is that the absolute aggregate and HCP clearance shift at commercial scale or over the resin life-cycle, or that the incoming aggregate load is higher than characterized; this is mitigated by defining the operating region on the qualified SDM, challenging the step against a representative worst-case aggregate feed, confirming the region at scale in Stage 2, and carrying the WC-CPP ranges forward with in-process monitoring and resin life-cycle control.

12 Approvals

Approval of this plan authorizes execution of the characterization described herein. Signatures are recorded in Table [1](#).

References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
- International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.
- International Council for Harmonisation. 2023. *ICH Q9(R1): Quality Risk Management*. ICH.
- International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.
- Parenteral Drug Association. 2013. *Technical Report No. 60: Process Validation — a Lifecycle Approach*. PDA.
- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned screening design

Coded two-level full-factorial design (factors A–D per Table 6.2; coded levels $-1/0/+1$). Responses will be recorded at execution and reported in PCR-007.

Table 12.1: Appendix A — planned screening design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Appendix B — Planned response-surface design

Coded face-centred central-composite design (factors A–D; coded levels $-1/0/+1$).

Table 12.2: Appendix B — planned response-surface design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0